Research Portfolio

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Current Work

Dust Growth and Vertical Settling in Protoplanetary Disks

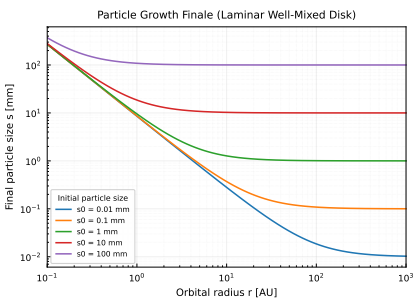
Institution: Indian Institute of Technology Roorkee
Supervisor: Dr. Akshay Bhatnagar
Duration: Jan 2026 – Present

Overview: Investigating the vertical evolution of dust in protoplanetary disks, focusing on the interplay between aerodynamic coupling, gravitational settling, turbulent diffusion, and particle growth.

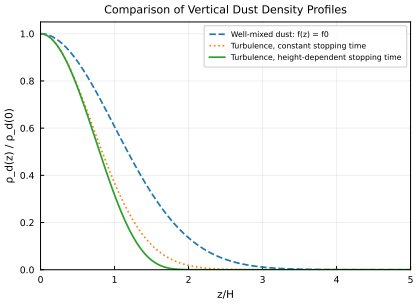
Methodology:

- Derived analytical solutions for dust growth during laminar vertical settling and steady-state dust distributions under competing settling and turbulent diffusion.
- Numerically integrated particle trajectories and grain-growth equations using Epstein drag, validating numerical results against analytical predictions and solving for vertical dust structures under different turbulent-diffusion prescriptions.
- Investigated how the settling-to-diffusion ratio and height-dependent Epstein drag influence dust concentration and vertical distribution across protoplanetary disks.
- Developing a vertically resolved, multi-species dust-coagulation framework coupling dust evolution with coagulation equations to investigate grain growth under turbulence and gas dynamics governed by the Navier–Stokes equations.

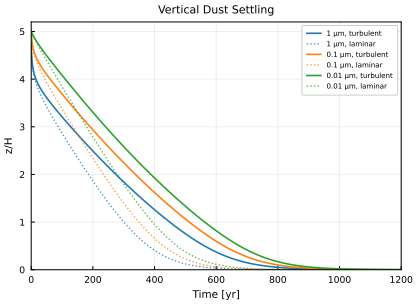
Key Results: The analytical and numerical calculations show that the vertical dust distribution depends on the competition between settling and turbulent diffusion. Accounting for the height dependence of the Epstein stopping time produces a more concentrated dust layer than the constant-stopping-time approximation. In the laminar model, the final grain size is determined by the dust column encountered during settling.



(a) final grain size vs. radius



(b) Dust distribution with diffusion



(c) Settling of dust under diffusion

Past Work

Time-series Analysis of Variable Stars (TESS and ARIES observations)

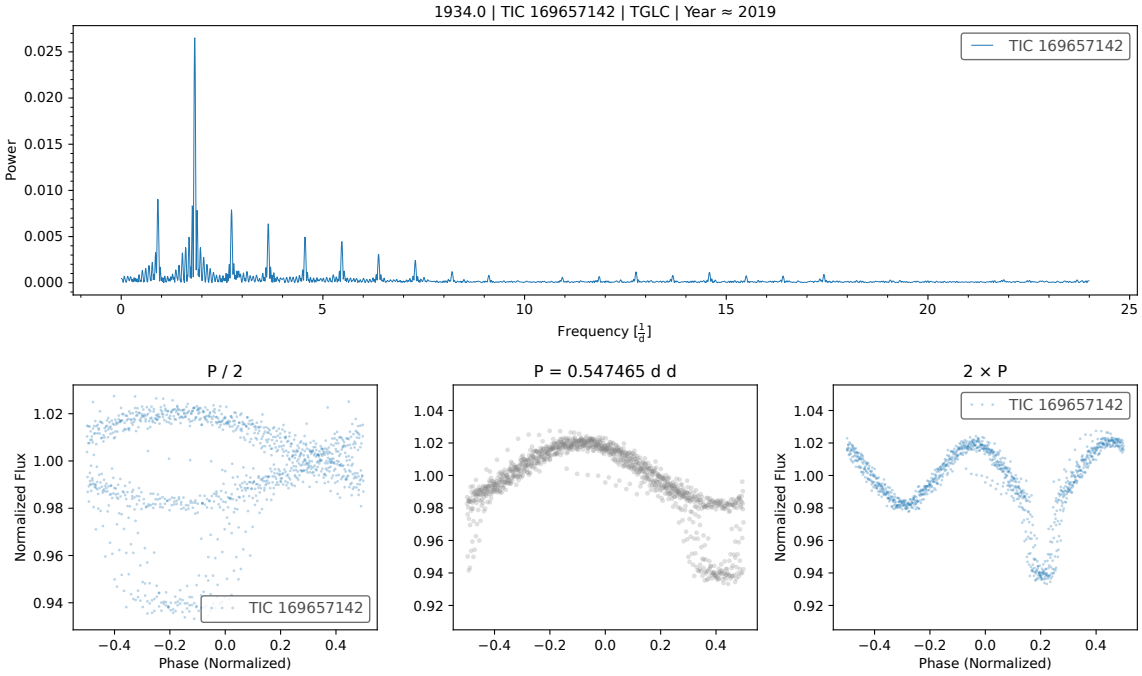
Institution: Aryabhatta Research Institute of Observational Sciences
Supervisor: Dr. Yogesh Chandra Joshi
Duration: Nov 2025 – Feb 2026

Overview: Analysis of photometric variability in open cluster NGC 2506 using long-term optical observations.

Methodology:

- Developed a pipeline to deal with ground based data collected by DOT-3.6m optical telescope at ARIES.
- Implemented Lomb–Scargle and Box Least Squares algorithms in aim to search for variability such as eclipsing binaries.
- Cross-matched results with TESS and Gaia DR3 catalogues

Key Outcome: Was able to classify numerous eclipsing binaries and δ-scuti stars as well as determine their time periods from short time (< 5 hrs) to very long scale (> 100 days).



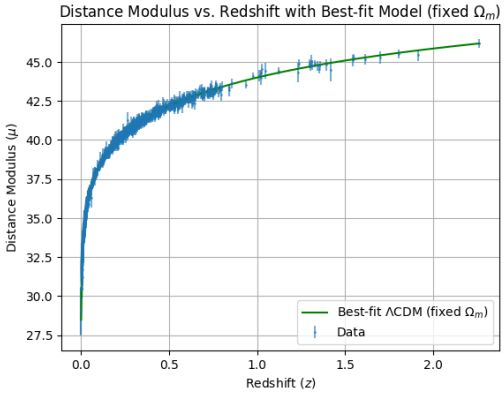
Example light curve showing periodic variability

Cosmological Parameter Estimation using Pantheon+ Dataset

Institution: India Space Academy
Duration: June 2025 – July 2025

Overview: Extracted best-fit cosmological parameters (H_0, Ω_m) via optimization (ΛCDM) and validated model consistency through residual analysis.

Key Outcome: Estimated the Hubble constant as $H_0 = 72.97 \pm 0.26$ km s^{-1} Mpc^{-1} and matter density $\Omega_m = 0.351 \pm 0.019$ through ΛCDM model fitting. Derived the age of the universe to be ~ 12.4 Gyr, consistent with late-time cosmological probes but exhibiting tension with Planck CMB measurements. Residual analysis showed no significant systematic bias, indicating good agreement between the model and observational data across redshift ranges.



Hubble diagram with fitted model